A Minor Project On

Design and Development of a Digital Twin Framework for Biologically Inspired Cognitive Robotic Systems

Presented By
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V B. Tech I Semester B. Tech+ M. Tech (IDP - AMS)



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INTRODUCTION

- Autonomous robotics demands real-time adaptive behaviour under noisy, dynamic physical environments — a challenge that rigid hand-coded controllers fail to meet.
- Biological organisms solve this through tightly-coupled neural, chemical, and bodily feedback loops operating across multiple temporal and spatial scales.
- Cognitive Autonomous Recurrent Lifeform (CARL) is proposed as a biologically-inspired digital organism, simulated within the MuJoCo physics engine, to replicate these multi-scale biological control principles.
- CARL integrates seven biologically-inspired mechanisms: an LSM reservoir brain, 4-factor endocrine neuromodulation (Dopamine, Serotonin, Norepinephrine, Acetylcholine), Spike-Timing-Dependent Plasticity (STDP), hippocampal grid/place cell navigation, a Free-Energy active inference engine, Physarum-style adaptive route reinforcement, and a mirror-neuron-based social learning layer.
- Traditional autonomous systems treat perception, navigation, memory, and motor control as isolated software modules — leading to brittle integration and failure under environmental perturbation.
- Biological brains maintain robustness through continuous prediction-error minimisation (Free-Energy Principle), neuromodulatory gain control, and activity-dependent synaptic reorganisation.

LITERATURE REVIEW — MODULE 1: PHYSICS & EMBODIMENT

Author(s)	Title	Methodology	Key Finding	Role in CARL
1. Todorov, Erez & Tassa (2012)	MuJoCo: A Physics Engine for Model-Based Control	Development of a contact-rich rigid-body physics engine optimized for model-based control and RL.	Provides fast, stable simulation of contact dynamics suited to real-time control-loop research.	Core simulation substrate — the digital twin environment CARL's body and physics run in.
2. Campion, Bastin & D'Andrea-Novel (1996)	Structural Properties and Classification of Kinematic and Dynamic Models of Wheeled Mobile Robots	Analytical classification of wheeled-robot kinematic/dynamic configurations and constraints.	Establishes a systematic framework for modelling mobility constraints across wheel configurations.	Grounds the differential-drive kinematic model used for CARL's physical embodiment.

LITERATURE REVIEW — MODULE 2: RESERVOIR BRAIN (LSM)

Author(s)	Title	Methodology	Key Finding	Role in CARL
3. Maass, Natschläger & Markram (2002)	Real-Time Computing Without Stable States	Theoretical and simulation analysis of fixed, randomly-connected recurrent spiking circuits with a linear readout.	A fixed 'liquid' reservoir supports real-time computation on time-varying input via a simple readout — no stable attractor states required.	Foundational theory for CARL's LSM reservoir brain.
4. Sussillo & Abbott (2009)	Generating Coherent Patterns of Activity from Chaotic Neural Networks	Introduces FORCE learning: a recursive least-squares (RLS) method for training readout weights of chaotic RNNs online.	Chaotic recurrent networks can be rapidly tamed into stable, complex output patterns via fast recursive weight updates.	Directly grounds the RLS-based online training of CARL's LSM readout layer.
5. Lukoševičius & Jaeger (2009)	Reservoir Computing Approaches to Recurrent Neural Network Training	Comprehensive survey of Echo State Network and Liquid State Machine training methods.	Reservoir computing separates fixed high-dimensional dynamics from a simple trainable readout, simplifying RNN training.	General survey grounding CARL's reservoir-based architecture choice.

LITERATURE REVIEW — MODULE 3: SPIKING NEURONS & STDP

Author(s)	Title	Methodology	Key Finding	Role in CARL
6. Song, Miller & Abbott (2000)	Competitive Hebbian Learning Through Spike-Timing-Dependent Synaptic Plasticity	Computational modelling of STDP as an unsupervised, competitive learning rule in spiking networks.	Asymmetric STDP time windows drive competition among synapses, selectively strengthening causally-consistent connections.	Core computational basis for CARL's STDP learning rule.
7. Burkitt (2006)	A Review of the Integrate-and-Fire Neuron Model I: Homogeneous Synaptic Input	Analytical review of the leaky integrate-and-fire (LIF) neuron model under noisy synaptic drive.	Establishes closed-form/statistical descriptions of LIF firing behaviour under stochastic input.	Grounds the LIF neuron model used inside CARL's LSM/spiking layers.
8. Bi & Poo (1998)	Synaptic Modifications in Cultured Hippocampal Neurons	Paired intracellular recordings measuring synaptic strength change as a function of pre/post spike timing.	First direct empirical evidence that potentiation/depression depends on millisecond-scale spike-timing order — the empirical origin of STDP.	Empirical/biological grounding validating CARL's STDP rule against real neural mechanisms.

LITERATURE REVIEW — MODULE 4: ENDOCRINE / NEUROMODULATION

Author(s)	Title	Methodology	Key Finding	Role in CARL
9. Aston-Jones & Cohen (2005)	An Integrative Theory of Locus Coeruleus-Norepinephrine Function	Synthesis of neurophysiological, pharmacological, and behavioural evidence on the LC-NE system.	Norepinephrine sets a global 'gain' on responsiveness, shifting behaviour between focused (exploitative) and labile (exploratory) modes.	Grounds CARL's norepinephrine-driven gain modulation of the LSM reservoir under threat/uncertainty.
10. Schultz, Dayan & Montague (1997)	A Predictive Reward Signal of Dopamine Neurons	Electrophysiological recording of dopamine neurons in monkeys during reward-predictive conditioning.	Dopamine neurons encode reward-prediction error, not reward itself — firing shifts to the earliest reliable predictor of reward.	Grounds CARL's dopamine signal as a reward-prediction-error / curiosity-driving mechanism.
11. Doya (2002)	Metalearning and Neuromodulation	Theoretical framework mapping dopamine, serotonin, norepinephrine, and acetylcholine onto distinct meta-learning parameters.	Each neuromodulator is proposed to tune a specific computational hyperparameter (prediction error, time-scale, gain, learning rate).	Directly grounds mapping CARL's 4-factor neuromodulation onto specific computational functions.
12. Robbins & Arnsten (2009)	The Neuropsychopharmacology of Fronto-Executive Function	Review of pharmacological/lesion studies on catecholamine and cholinergic effects on prefrontal executive function.	Moderate catecholamine/acetylcholine levels optimize attention and working memory in an inverted-U relationship.	Grounds CARL's acetylcholine-gated attention/learning-readiness mechanism.

LITERATURE REVIEW — MODULE 5: SPATIAL NAVIGATION (HIPPOCAMPUS)

Author(s)	Title	Methodology	Key Finding	Role in CARL
13. Moser, Kropff & Moser (2008)	Place Cells, Grid Cells, and the Brain's Spatial Representation System	Review of electrophysiological recordings of hippocampal place cells and entorhinal grid cells in freely-moving rodents.	Grid cells provide a path-integration-based metric coordinate system; place cells combine this with sensory context.	Core grounding for CARL's hippocampal navigation mechanism.
14. O'Keefe & Dostrovsky (1971)	The Hippocampus as a Spatial Map	Single-unit recordings from hippocampal neurons in freely-moving rats.	First discovery of 'place cells' — neurons firing selectively based on the animal's location in space.	Foundational/historical grounding for CARL's location-tagged memory mechanism.
15. Burak & Fiete (2009)	Accurate Path Integration in Continuous Attractor Network Models of Grid Cells	Computational modelling of grid-cell firing using continuous attractor networks driven by velocity input.	A continuous attractor network path-integrates accurately over long trajectories using local velocity updates.	Provides the computational implementation model for CARL's grid-cell path integration.

LITERATURE REVIEW — MODULE 6: PATH PLANNING & PHYSARUM NETWORK DIFFUSION

Author(s)	Title	Methodology	Key Finding	Role in CARL
16. Hart, Nilsson & Raphael (1968)	A Formal Basis for the Heuristic Determination of Minimum Cost Paths (A*)	Formal algorithmic derivation and proof of optimality/admissibility conditions for heuristic graph search.	A* guarantees shortest-path optimality when its heuristic never overestimates true cost, while remaining efficient.	Classical path-planning baseline against which CARL's memory-guided route selection is compared.
17. Tero et al. (2010)	Rules for Biologically Inspired Adaptive Network Design	Empirical Physarum polycephalum growth experiments across food-source layouts, combined with a mathematical tube-adaptation model.	Local reinforcement of high-flow tubes and decay of low-flow tubes yields efficient, fault-tolerant networks without central planning.	Grounds CARL's Physarum-style reinforcement of frequently-used safe routes across the hippocampal map.

LITERATURE REVIEW — MODULE 7: MOTOR CONTROL & LEARNING ALGORITHMS

Author(s)	Title	Methodology	Key Finding	Role in CARL
18. Borenstein & Koren (1991)	The Vector Field Histogram — Fast Obstacle Avoidance for Mobile Robots	Real-time obstacle avoidance using a polar histogram of sensor-detected obstacle density around the robot.	VFH enables fast, smooth real-time avoidance without full path replanning.	Classical reactive-control baseline complementing CARL's LSM/SNN reflex layer.
19. Brown, Kairiss & Keenan (1990)	Hebbian Synapses: Biophysical Mechanisms and Algorithms	Review of biophysical LTP/LTD induction mechanisms underlying Hebbian synaptic plasticity.	Synaptic strength changes as a function of correlated pre- and post-synaptic activity, the biological basis for associative learning.	General theoretical foundation underlying CARL's STDP mechanism.
20. Hasani, Lechner, Amini, Rus & Grosu (2021)	Liquid Time-Constant Networks	Continuous-time neural ODE formulation with input-dependent, learnable per-neuron time constants.	Adaptive per-neuron time constants match processing speed to input dynamics, with far fewer parameters than LSTMs.	Candidate upgrade path for CARL's fixed-random LSM reservoir toward adaptive dynamics.

LITERATURE REVIEW — MODULE 8: THEORETICAL FOUNDATIONS (I)

Author(s)	Title	Methodology	Key Finding	Role in CARL
21. Rahimi et al. (Kanerva-style HDC)	Hyperdimensional Computing for Efficient Embodied Learning and Memory	Algebraic analysis and robotic benchmarking of high-dimensional (~40,000-D) bipolar vector representations.	Near-orthogonal high-dimensional vectors give constant-time binding, bundling, and noise-robust associative retrieval.	Candidate implementation for CARL's hippocampal location-outcome binding, robust to sensor noise.
22. Sterling (2020)	The Concept of Allostasis and Predictive Energy Allocation in Neural Systems	Theoretical synthesis of physiological/neuroscientific evidence on predictive drive regulation.	Biological systems regulate internal states predictively (allostasis) rather than only reactively (homeostasis).	Grounds tying CARL's neuromodulation/drive system to predictive, free-energy-based regulation.
23. Friston (2010)	The Free-Energy Principle: A Unified Brain Theory?	Theoretical/mathematical derivation unifying perception, learning, and action under variational free-energy minimisation.	Agents act and update beliefs to minimise long-run prediction error (surprise), unifying perception and action.	Direct theoretical grounding for CARL's free-energy active inference decision layer.

LITERATURE REVIEW — MODULE 9: THEORETICAL FOUNDATIONS (II)

Author(s)	Title	Methodology	Key Finding	Role in CARL
24. Buzsáki (2002)	Theta Oscillations in the Hippocampus	Review of electrophysiological studies of hippocampal theta-rhythm oscillations during navigation and memory tasks.	Theta oscillations temporally organise place-cell firing sequences, enabling encoding and replay of spatial trajectories.	Grounds the timing/sequencing mechanism for CARL's hippocampal trajectory encoding and replay.
25. Rizzolatti & Craighero (2004)	The Mirror-Neuron System	Review of electrophysiological (macaque) and neuroimaging (human) evidence for neurons active during both self-performed and observed action.	A dedicated neural system maps observed actions of others onto one's own motor repertoire, supporting imitation.	Theoretical grounding for CARL's planned multi-agent mirror-neuron imitation layer (Phase 3 roadmap).

LITERATURE SUMMARY

- 25 papers were selected from a working base of 40, organised across 9 modules spanning physics/embodiment, reservoir computing, spiking plasticity, neuromodulation, spatial cognition, path planning, motor control, and theoretical foundations — covering every mechanism proposed for CARL.
- Maass (LSM theory) and Sussillo & Abbott (FORCE/RLS) ground CARL's reservoir brain and its online readout training.
- Song/Miller/Abbott and Bi & Poo ground CARL's STDP rule computationally and empirically; Burkitt grounds its LIF neuron model.
- Aston-Jones & Cohen, Schultz et al., Doya, and the fronto-executive review ground CARL's 4-factor neuromodulation (NE gain, DA reward-prediction, ACh attention-gating).
- Moser et al., O'Keefe & Dostrovsky, and Burak & Fiete ground CARL's hippocampal grid/place-cell navigation; Buzsáki grounds its theta-timed sequence replay.
- Tero et al. grounds CARL's Physarum-style route-network reinforcement; Hart/Nilsson/Raphael (A*) and Borenstein & Koren (VFH) provide classical planning/reflex baselines.
- Friston and Sterling ground CARL's free-energy active inference and predictive (allostatic) drive regulation.
- Rizzolatti & Craighero grounds the mirror-neuron imitation layer planned for CARL's multi-agent roadmap phase.

RESEARCH GAP

Within the literature surveyed for this project:

- No single reviewed architecture integrates a spiking/reservoir controller with a full deep-learning-style control loop within one real-time system.
- Limited work couples multi-factor endocrine neuromodulation (DA/NE/5-HT/ACh) with Liquid State Machine reservoir brains for continuous autonomous control.
- No reviewed work directly couples hippocampal continuous-attractor grid cells with a real-time MuJoCo physics-based motor controller for embodied navigation.

PROPOSED GAP

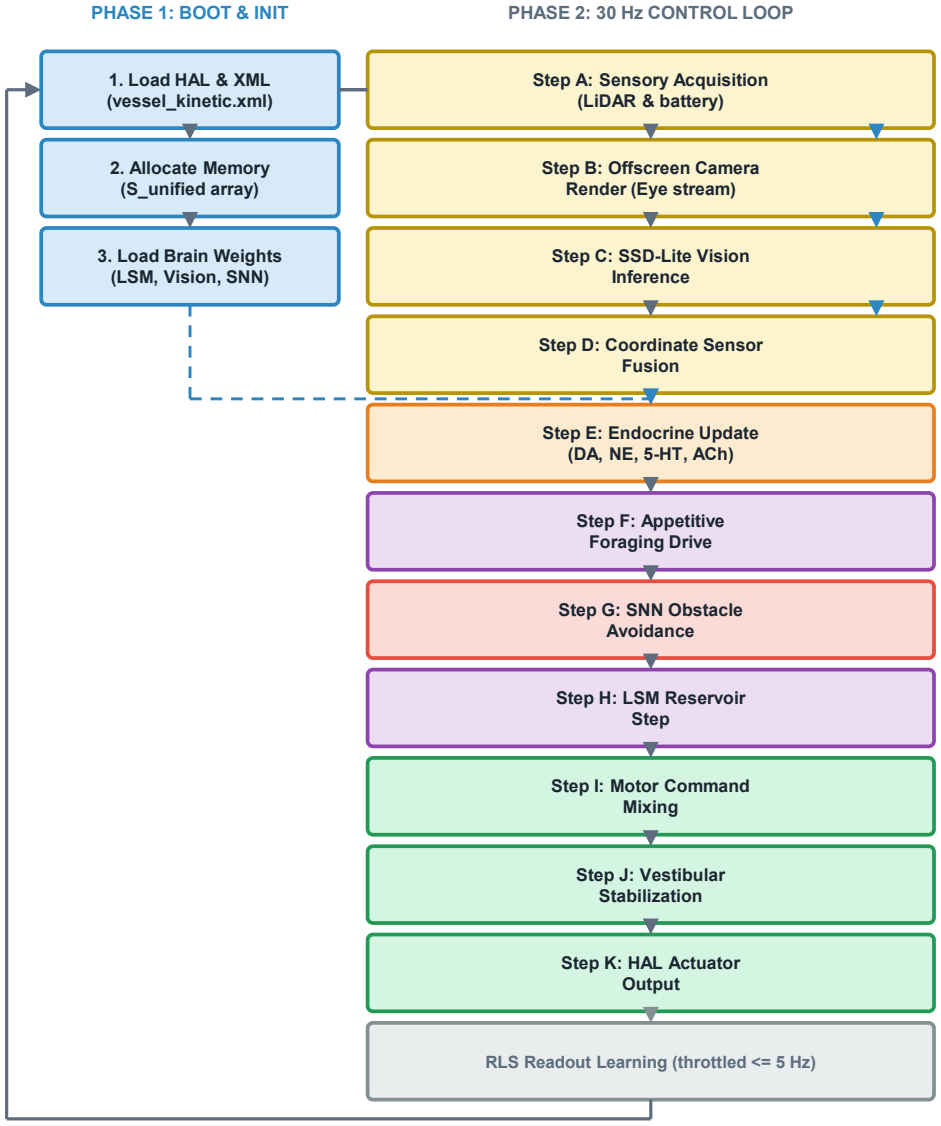
- Develop CARL as an integrated architecture combining LSM reservoir computing, STDP spiking plasticity, 4-factor endocrine gain control, hippocampal grid-cell navigation, Free-Energy active inference, and Physarum-style route reinforcement within a single 30 Hz real-time MuJoCo simulation loop.
- Validate the architecture through systematic digital twin experiments measuring adaptive behaviour emergence, energy homeostasis, and spatial navigation accuracy.

OBJECTIVES

- To design and simulate CARL as an integrated cognitive-robotic architecture within the MuJoCo physics engine, replicating multi-scale biological control principles.
- To integrate a Liquid State Machine (LSM) reservoir brain with 4-factor endocrine neuromodulation (Dopamine, Serotonin, Norepinephrine, Acetylcholine) for adaptive gain control.
- To implement Spike-Timing-Dependent Plasticity (STDP) for activity-dependent synaptic reorganisation and continual learning.
- To develop hippocampal grid/place cell-based spatial navigation for embodied wayfinding.
- To incorporate a Free-Energy active inference engine for continuous prediction-error minimisation and action selection.
- To extend the route network with Physarum-inspired adaptive reinforcement, strengthening frequently-used safe paths across the explored map.
- To establish a mirror-neuron-based imitation layer as a long-term roadmap item for multi-agent, socially-learned behaviour.
- To validate the architecture through digital twin experiments measuring adaptive behaviour emergence, energy homeostasis, and spatial navigation accuracy within a real-time (30 Hz) control loop.

METHODOLOGY

1. SYSTEM ARCHITECTURE: GLOBAL COGNITIVE STACK



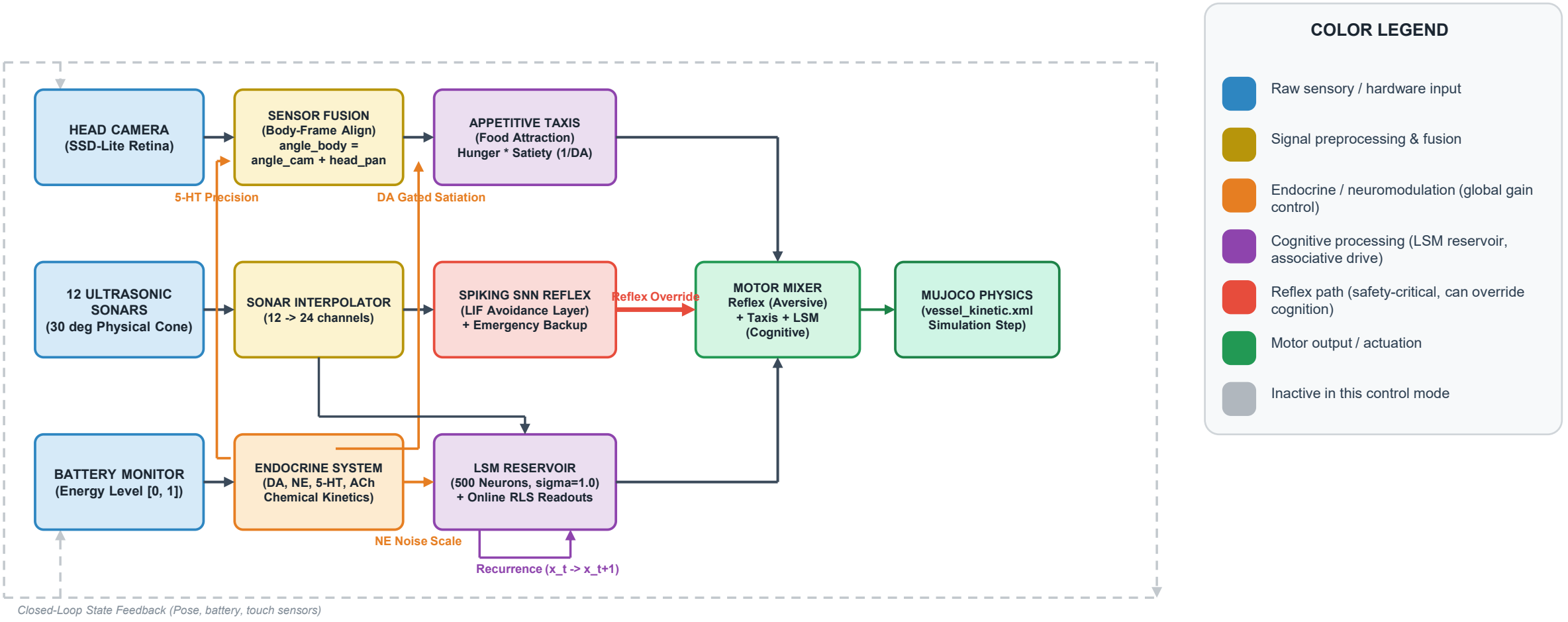
COLOR LEGEND

- Raw sensory / hardware input
- Signal preprocessing & fusion
- Endocrine / neuromodulation (global gain control)
- Cognitive processing (LSM reservoir, associative drive)
- Reflex path (safety-critical, can override cognition)
- Motor output / actuation
- Auxiliary / logging block (RLS readout)

Two-phase execution: Boot & Initialization (Phase 1) feeding a closed-loop 30 Hz sensorimotor control cycle (Phase 2).

METHODOLOGY (CONTINUED)

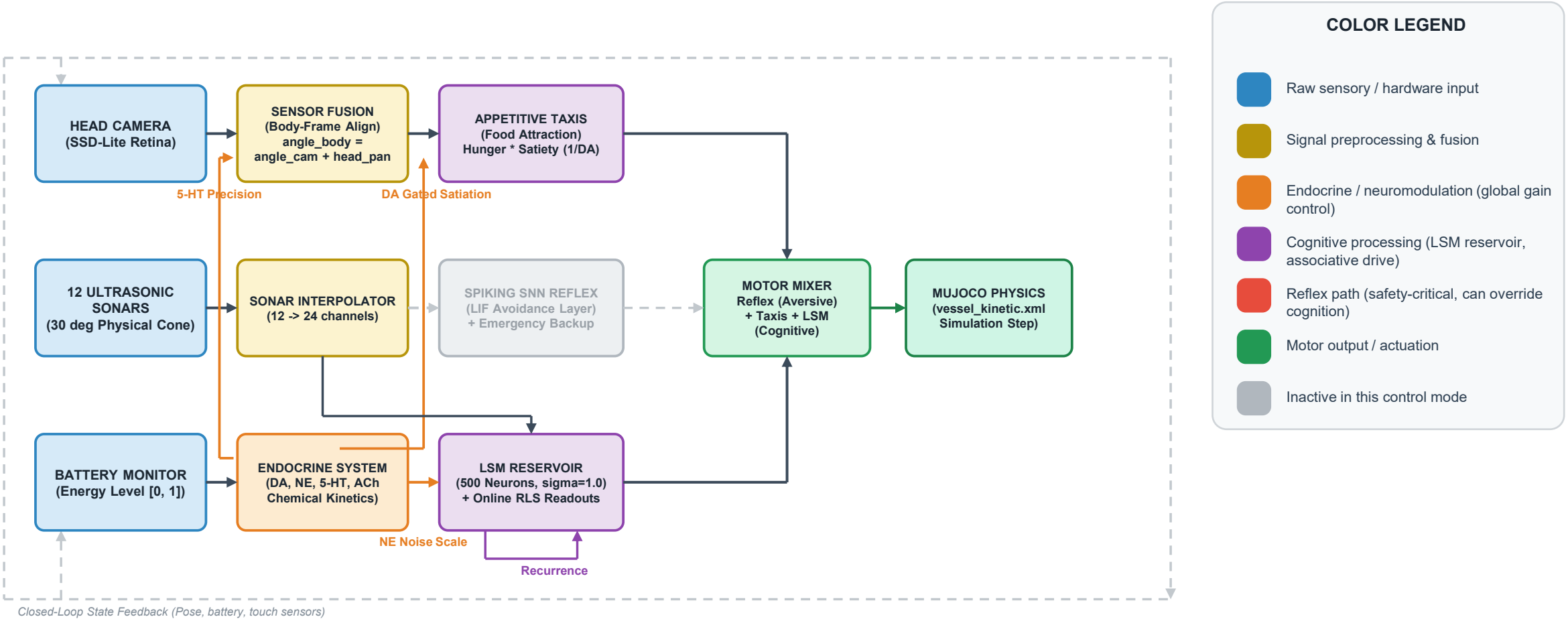
2. COMPLETE CONTROL LOOP & ENDOCRINE MODULATION PATHWAYS



Full pathway view showing 5-HT precision gating, DA-gated satiation, NE noise-scale modulation, and LSM recurrence feeding the Motor Mixer.

METHODOLOGY (CONTINUED)

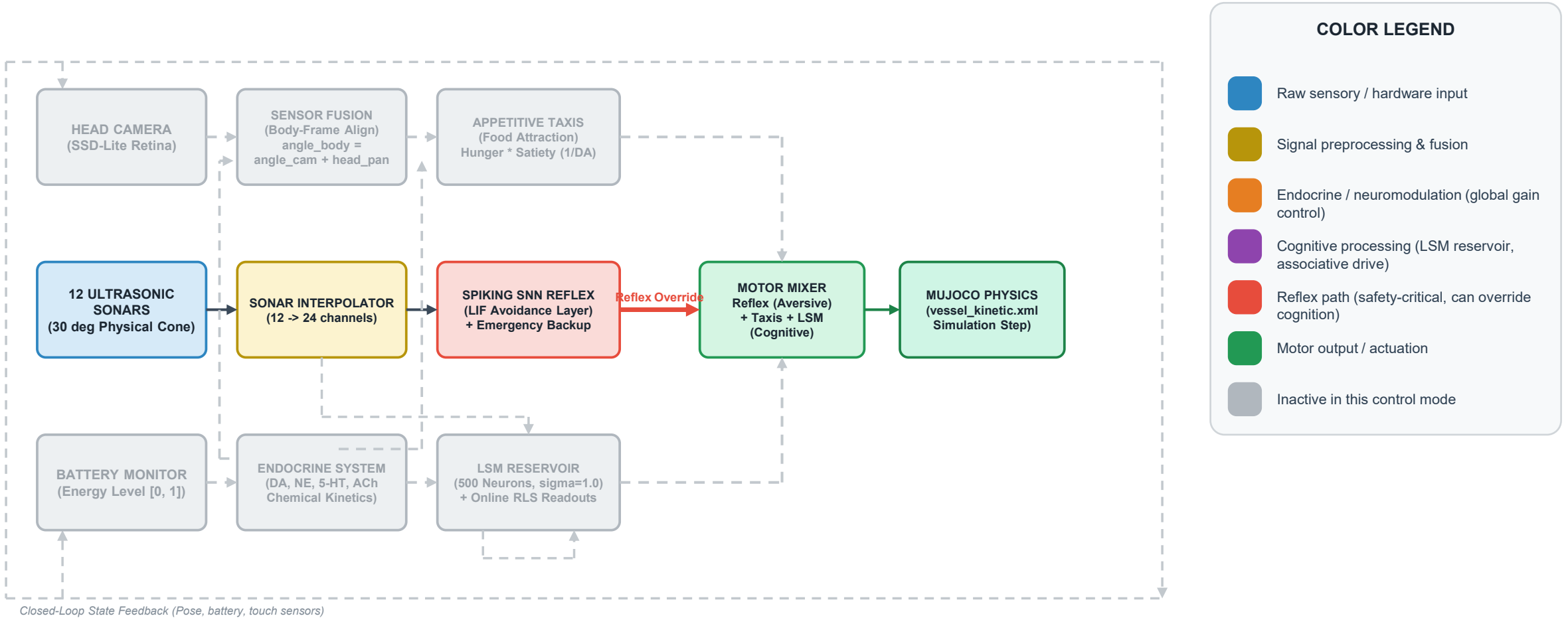
3. CONTROL LOGIC FLOW: NORMAL MODE



In the absence of an emergency trigger, sensor fusion, appetitive taxis, endocrine state, and the LSM reservoir jointly drive the Motor Mixer.

METHODOLOGY (CONTINUED)

4. CONTROL LOGIC FLOW: REFLEX MODE



Under an imminent collision/hazard signal, the Spiking SNN reflex layer overrides cognitive (LSM) output directly at the Motor Mixer.

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THANK YOU